

ISTANBUL ATATURK, Turkey

WMO: 170600

Lat: 40.967N

Lon: 28.817E

Elev: 33

StdP: 100.93

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.4	0.0	-6.6	2.2	2.2	-5.0	2.5	2.8	13.5	1.7	12.4	2.8	6.5	0	0.505

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.2	32.1	21.5	30.9	21.3	29.9	21.1	24.5	28.0	23.6	27.6	22.9	27.2	5.6	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.2	18.1	26.5	22.2	17.0	26.0	21.5	16.2	25.7	74.3	27.8	70.9	27.4	68.2	27.3	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.2	10.0	9.0	DB	-4.0	34.9	2.4	1.6	-5.7	36.1	-7.1	37.0	-8.4	37.9	-10.1	39.0	
			WB	-4.8	25.2	2.2	1.2	-6.4	26.0	-7.8	26.8	-9.0	27.5	-10.6	28.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	6.5	7.0	9.0	13.1	18.2	22.9	25.7	25.9	21.9	17.1	12.6	8.5
	DBStd	7.61	3.69	3.64	3.51	3.43	2.95	2.64	1.97	2.11	2.81	3.05	3.59	3.63
	HDD10.0	374	118	94	62	9	0	0	0	0	0	1	17	73
	HDD18.3	1716	367	316	289	160	39	1	0	0	4	59	174	306
	CDD10.0	2474	9	11	31	103	254	388	488	494	357	220	94	26
	CDD18.3	776	0	0	0	3	34	139	229	236	112	21	2	0
	CDH23.3	5539	0	0	0	13	131	858	1952	2018	531	36	1	0
	CDH26.7	1507	0	0	0	1	17	208	589	602	87	2	0	0
Wind	WSAvg	4.5	4.8	4.8	4.4	4.0	3.9	4.2	4.8	4.9	4.3	4.3	4.2	4.9
Precipitation	PrecAvg	640	95	62	58	51	30	28	22	31	57	85	113	
	PrecMax	1026	193	138	132	166	115	59	85	94	162	139	208	361
	PrecMin	452	12	12	14	11	3	0	0	0	1	2	35	21
	PrecStd	126	52	33	35	34	24	19	20	24	33	37	41	69
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.9	16.9	20.1	25.1	28.8	32.2	34.0	33.9	30.9	26.2	22.2	18.8
		MCWB	11.0	11.1	12.2	15.4	18.5	20.8	20.7	21.5	20.3	18.9	15.2	13.2
	2%	DB	14.1	14.9	17.8	22.1	26.2	30.4	32.0	32.1	28.9	24.1	20.2	16.1
		MCWB	10.2	10.3	11.6	14.4	17.4	20.2	21.3	22.0	19.8	18.3	15.0	12.4
	5%	DB	12.9	13.2	15.9	20.1	24.8	29.0	30.9	30.9	27.2	22.8	18.9	14.9
		MCWB	9.7	9.7	11.0	13.6	17.1	20.0	21.2	21.8	19.8	17.6	14.6	11.7
	10%	DB	11.8	12.0	14.1	18.6	23.1	27.7	29.8	29.8	26.1	21.2	17.4	13.6
		MCWB	9.0	9.1	10.1	12.8	16.5	19.5	21.1	21.6	19.4	17.1	13.9	10.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.9	12.0	13.3	16.3	20.2	23.3	25.5	25.7	23.2	20.7	17.2	14.1
		MCDB	14.5	15.4	18.2	22.5	25.4	27.7	28.1	28.7	27.3	23.8	19.9	17.2
	2%	WB	10.8	10.9	12.2	15.1	18.9	22.4	24.5	24.7	22.1	19.5	16.0	12.8
		MCDB	13.2	13.9	16.4	20.3	23.6	26.7	27.7	28.3	25.8	22.4	19.0	15.6
	5%	WB	10.0	10.1	11.3	14.2	18.1	21.7	23.6	23.9	21.2	18.5	15.0	11.9
		MCDB	12.2	12.8	15.1	19.1	22.8	26.1	27.4	28.0	25.0	21.6	17.9	14.5
	10%	WB	9.2	9.3	10.5	13.3	17.4	20.9	22.9	23.2	20.7	17.6	14.2	11.0
		MCDB	11.6	11.9	13.8	17.5	21.9	25.6	27.2	27.3	24.5	20.7	17.0	13.3
Mean Daily Temperature Range	5% DB	MDBR	4.7	5.3	6.2	7.4	7.6	7.7	7.6	7.2	6.9	5.8	5.4	4.7
		MCDBR	6.0	6.9	8.8	10.2	9.8	9.4	8.9	8.5	8.1	7.3	6.8	5.3
		MCWBR	3.7	4.1	4.4	4.7	4.4	4.0	3.9	3.3	3.4	3.5	3.5	3.5
	5% WB	MCDBR	5.5	6.5	7.8	9.1	8.4	8.1	7.7	7.3	7.3	6.6	6.0	5.1
		MCWBR	3.9	4.2	4.4	4.7	4.3	4.0	3.9	3.3	3.6	3.6	3.6	3.7
Clear-Sky Solar Irradiance	taub	0.372	0.403	0.439	0.480	0.443	0.433	0.436	0.445	0.434	0.415	0.385	0.376	
	taud	2.313	2.206	2.123	2.028	2.209	2.282	2.285	2.260	2.274	2.294	2.324	2.306	
	Ebn at Noon	759	789	803	797	838	846	839	820	801	768	738	718	
	Edh at Noon	91	116	141	166	141	132	130	130	120	106	89	85	
All-Sky Solar Radiation	RadAvg	1.48	2.16	3.46	4.85	6.18	6.98	7.17	6.19	4.57	2.97	1.93	1.35	
	RadStd	0.22	0.32	0.33	0.53	0.46	0.52	0.24	0.32	0.33	0.27	0.26	0.20	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.75	N/A	N/A	+0.59	N/A	N/A	-53	-141	+171	+85
(47)	Regional (13 neighbors)	+0.52	N/A	N/A	N/A	N/A	N/A	N/A	-98	N/A	+44

Nomenclature: See separate page